



How to Size an Expansion Tank

Three things, in this order: find out if a tank is even needed, collect nine numbers, type them in and read the answer. Step one can save you the whole job.

OPEN THE CALCULATOR

Double-click **NFPA13_Expansion_Tank_Calculator.html**. It opens in your web browser. It works on a phone. It does not need internet.

PART 1 Do you even need a tank?

Answer three questions first. Any one of them can end this early.

Question 1 — Is there a backflow preventer?

Look at where the antifreeze system connects to the building's water supply. Is there a backflow prevention device in that connection?

- **Yes, there is one.** Keep going. You probably need a tank.
- **No, there isn't one.** Stop. A tank is usually not required here.

WHY THIS MATTERS

An expansion tank exists because the antifreeze has nowhere to go. A backflow preventer is what blocks the escape route. No blocker, no trapped pressure, no tank.

Question 2 — How much antifreeze is in the system?

CHECK THIS BEFORE YOU PROPOSE A TANK

40 gallons or less? A listed 1/2 inch relief valve is allowed *instead* of a tank, and costs a fraction as much. If you propose a tank when a relief valve would do, you may lose the job to someone who checked.

More than 40 gallons? Keep going. You need a tank. You'll measure this properly in Part 2 — a rough guess is fine for now.

Question 3 — Is there an automatic pump keeping pressure up?

This is rare. You'd know it if you saw it — a pump and controls that hold pressure on the antifreeze side, dump fluid when it gets hot, and add fluid back when it cools. If all three of those exist, **stop**: the tank rules don't apply.

PART 2 Go collect nine numbers

Take a clipboard. There's a blank sheet at the end of this guide.

Measure. Do not assume. Every number on this list has been wrong on somebody's tag at some point.

NO. 1 What antifreeze is in there?

WHAT YOU NEED

The brand and the exact product name.

HOW TO GET IT

Read the placard on the antifreeze system main valve. Then go look at the drum or the paperwork if you can, because tags get old.

WHY IT MATTERS

This is the single biggest thing. Different antifreezes swell by different amounts when they warm up. Two products from the *same* company, with almost the same name, needed a 17.5 gallon tank and a 53 gallon tank on the identical system. You cannot guess this from the label.

WATCH OUT FOR

A system that just says "antifreeze" and nothing else. If you can't identify the product, you cannot size the tank. Find out first.

NO. 2 How strong is it?

WHAT YOU NEED

The concentration, as a percent.

HOW TO GET IT

Measure it with a hydrometer or a refractometer. Take a sample at the supply inlet **and** at a far point in the system. Write down both.

WHY IT MATTERS

A 30% mix and a 48% mix are different liquids. They swell by different amounts. The percent changes the tank size.

WATCH OUT FOR

- Someone topped the system off with water years ago. The tag still says the original number. Your meter tells the truth.
- If the two samples don't match each other, the system isn't mixed evenly. That's a separate problem — report it.
- Some modern antifreezes come pre-mixed at 100% and must never be diluted. If it's one of those, there's nothing to measure.

NO. 3 How much liquid is in the pipes?

WHAT YOU NEED

Total gallons of antifreeze in the system. **Do not count the expansion tank itself** — just the pipe.

HOW TO GET IT — WAY A

Walk the system and add it up. Write down every run: size, type of pipe, how many feet. The calculator does the arithmetic.

HOW TO GET IT — WAY B

If there's a reliable record, use it and type the number straight in.

WHY IT MATTERS

More liquid means more swelling means a bigger tank. It scales straight across — double the pipe, double the swelling.

WATCH OUT FOR

- **Old drawings lie.** Somebody added a branch in 2009 and didn't redline it. Walk it.
- **Thin-wall pipe is not Schedule 10.** Super-Flo, Mega-Flo and Eddy-Flow are thinner and hold a different amount. Get the real numbers from that pipe's data sheet — don't pick "Schedule 10" and hope.

NO. 4 How cold will the pipes get?

WHAT YOU NEED

The lowest temperature the antifreeze will ever see, in °F.

HOW TO GET IT

Think about where the pipe actually runs. An unheated attic in January. A loading dock. A freezer. Use the real local design low.

WHY IT MATTERS

This is where the liquid is densest and packed tightest. It's the starting point for the swelling.

WATCH OUT FOR

This is *not* the antifreeze's freeze point. The freeze point is a property of the liquid. This is how cold the space gets.

NO. 5 How hot will the pipes get?

WHAT YOU NEED

The highest temperature the antifreeze will ever see, in °F.

HOW TO GET IT

For an attic or a concealed space above a ceiling, the answer is usually **140°F to 150°F**. Summer sun on a roof deck gets brutal. If you think it's lower, be able to say why.

WHY IT MATTERS

This is the number that drives the whole answer. This is where the liquid swells the most and the tank has to swallow it.

WATCH OUT FOR

Guessing low. Put in 100°F when the attic really hits 150°F and you'll size a tank far too small that looks fine on paper. The calculator warns you below 140°F. Take that warning seriously.

NO. 6 How warm is the air when the tank gets charged?

WHAT YOU NEED

The temperature, in °F, at the moment air goes into the tank — **before** it gets installed.

HOW TO GET IT

Usually shop temperature or the back of the truck. Around 70°F is typical. Charging it in an unheated warehouse in February? It is not 70°F.

WHY IT MATTERS

The code specifically requires this in the math. Air behaves differently at different temperatures, and the tank was charged somewhere other than where it lives.

NO. 7 What air pressure goes into the tank?

WHAT YOU NEED

The precharge pressure, in psi. This is a number you *set*, not one you find.

THE RULE FROM THE TANK MAKER

- Set it about **2 psi below** the water pressure at the tank location.
- **Never** set it below **25%** of the system's maximum rated pressure.

THIS ONE CATCHES PEOPLE

The pressure the tank ships with (often 25 psi) is a **shipping** charge, not the installed pressure. Type in the shipping number and the calculator thinks the tank holds more liquid than it really can — and you size it too small.

NO. 8 What's the water pressure at the tank?

WHAT YOU NEED

Static pressure, in psi, on the **water supply side** of the backflow preventer.

HOW TO GET IT

Put a gauge on it. Read it with no flow.

WHY IT MATTERS

This sets how much water gets pushed into the tank before anything even warms up.

NO. 9 What's the weakest part rated for?

WHAT YOU NEED

The maximum pressure the antifreeze system is allowed to reach, in psi.

HOW TO GET IT

Find the **lowest-rated component** in the antifreeze portion. Check every one: pipe, fittings, valves, sprinklers, the backflow device, gauges. The lowest number wins.

WHY IT MATTERS

The tank has to keep pressure under the weakest link. If one old valve is rated 150 psi, your answer is 150 — not 175.

WATCH OUT FOR

Defaulting to 175 psi because that's typical. On an existing system, go look. Older equipment is often lower.

PART 3 Type it in

Open the calculator. Work top to bottom.

- 1 **Pick the code edition.** Three buttons at the top: 2016, 2019, 2025. Pick the one your local inspector enforces. If you don't know, ask — don't guess. It changes which section numbers print on your paperwork.
- 2 **Enter the system volume.** Click *Pipe takeoff* if you walked the pipe — pick size, pick type, type the feet, then + *Add pipe run* for the next one. Click *Enter directly* if you already know the gallons.
- 3 **Pick the antifreeze.** If your product is in the dropdown, pick it and the densities fill in automatically from the manufacturer's data sheet. If it isn't, pick "Enter D_L / D_H by hand." If the calculator blocks your product, it turns red and tells you why — read it. That means a real problem, not a broken tool.
- 4 **Enter the densities** (only if entering by hand). D_L is the density at your coldest temperature. D_H is the density at your hottest. Both come off the data sheet.
- 5 **Enter the three temperatures.** T_0 is when the tank got charged. T_1 is coldest. T_2 is hottest.
- 6 **Enter the three pressures.** All three are gauge pressure — normal pressure off a gauge. The calculator does the conversion itself. P_0 is air into the tank, P_1 is water at the tank, P_2 is the weakest part.
- 7 **Press Calculate.** If you get a red box instead of an answer, something you typed can't be right and it tells you exactly what. Fix it and press again.

THREE RULES FOR THE DENSITY BOXES

- **D_L is always the bigger number.** Cold liquid is denser. If yours isn't, you've swapped them.
- **Use the same kind of number for both.** If one is g/mL, the other must be too. Don't mix a density with a specific gravity.
- **Read at YOUR two temperatures** — not the middle of the table. This is the most expensive mistake on the page. On a real case, grabbing a middle row produced a tank three sizes too small with no warning.

THE CALCULATOR WILL NOT GUESS FOR YOU

It refuses to produce a number it doesn't believe. A red box is the tool working, not failing.

PART 4 Read the answer

You get **two numbers**, and a tank must pass **both**. This is the part people get wrong.

Number A — Total chamber volume

How big the tank has to be overall.

Number B — Acceptance volume

How much liquid the tank can actually swallow.

These are not the same thing, and the second one is smaller. A tank has a rubber diaphragm inside. The liquid only fills part of the shell — the diaphragm stops it. So a 34 gallon tank might only accept 11 gallons of liquid.

Here’s the trap

TANK	SHELL SIZE	LIQUID IT CAN ACCEPT
FPT-42V	17.5 gal	11.0 gal
FPT-60V	25.0 gal	11.0 gal
FPT-70V	34.0 gal	11.0 gal

Three different tanks. Twice the shell. **Same 11 gallons of liquid.** Buying a bigger tank buys you nothing past that point. If you only checked shell size, you’d pick the 34 gallon tank, feel good, and install something that can’t take the fluid.

THE CALCULATOR DOES BOTH CHECKS

Every model gets a **Needs** column — how much that particular tank would have to swallow — and a Pass or Fail. Take the one marked **Smallest fit**.

IF IT SAYS “BIGGER IS NOT SAFER HERE”

Believe it. A bigger shell holds more water before anything even warms up, so a larger tank can fail where a smaller one passed. **Never substitute up without re-running this.**

Print it

Click **Print calc sheet for submittal**. You get a clean page with every input, every formula, every step, and the code sections. That’s what the inspector wants to see — the math, not just the answer.

PART 5 Before it goes in the proposal

- ☐ Concentration was **measured**, not read off a tag
- ☐ System volume came from **walking the pipe**, not old drawings
- ☐ Hottest temperature is realistic for that space (140–150°F for an attic)
- ☐ P_2 is the **lowest-rated component**, verified by looking
- ☐ Precharge is the **installed** pressure, not the shipping pressure
- ☐ Selected tank passes **both** checks
- ☐ Tank is **compatible** with this antifreeze — check the data sheet for required diaphragm material
- ☐ Tank pressure rating is at or above your P_2
- ☐ Calc sheet printed and attached
- ☐ Antifreeze data sheet attached — the one the densities came from

Mention these too, because someone will ask

- **Over 40 gallons?** Friction loss also has to be calculated a second way (Darcy–Weisbach). That's a hydraulic recalculation, not part of this tank sizing.
- **The placard requirements changed in 2025.** Older editions want the concentration printed on it; the 2025 edition wants the minimum-use temperature instead. Same section number, different information. Don't order placards off an old template.
- **Concentration has to be checked every year**, at two points.

ONE MORE THING WORTH RAISING

If this system contains **unlisted** glycerine or propylene glycol, it's only allowed to stay in service if the local inspector has looked at it and approved it to remain. If that approval doesn't exist, adding a tank may be money spent on a system that has to be changed anyway. Worth a conversation before the proposal goes out.

Field Sheet

Tear this off and take it with you.

Job	Date	Tech
_____	_____	_____

First, three yes/no questions

QUESTION	YES	NO
Backflow preventer in the connection?	☐	☐ ← STOP
System over 40 gallons?	☐	☐ ← relief valve?
Automatic pressure pump system?	☐ ← STOP	☐

Then, nine numbers

#	WHAT	VALUE	WHERE IT CAME FROM
1	Antifreeze brand + product		
2	Concentration at inlet	%	
2b	Concentration at far point	%	
3	System volume (no tank)	gal	
4	Coldest pipe temp — T_1	°F	
5	Hottest pipe temp — T_2	°F	
6	Temp at precharge — T_0	°F	
7	Precharge pressure — P_0	psi	
8	Static at tank — P_1	psi	
9	Lowest-rated component — P_2	psi	which part:

Pipe takeoff

SIZE	TYPE	FEET

Answer

Total volume needed	Acceptance needed
Tank selected	
Tank shell	Tank acceptance rating
Passes both checks? ☐ Yes ☐ No	